MIL-D-80229A
10 December 1986
SUPERSEDING
MIL-D-80229
19 October 1971

MILITARY SPECIFICATION

DEPOSITION MACHINE, CARBIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically operated, portable, tungsten carbide deposition machines, designed for manual or automatic application of tungsten carbide coatings to tool cutting edges and other high wear surfaces. The carbide deposit is applied by lightly rubbing an electrically charged, vibrating tungsten carbide electrode over the surface to be coated.

1.2 Classification. The depostion machines shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size 2 - 0.0001 to 0.002 inch, deposit thickness.
Size 3 - 0.0001 to 0.004 inch, deposit thickness.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of applying carbide coatings of various thickness to tool cutting edges and other high wear surfaces in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

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3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Ch XVII, Part 1910 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to

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perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components.

3.4.1 Power supply and control console. The power supply and operating controls shall be housed in one cabinet, with controls mounted on the front panel. The output of the power supply shall be direct current (DC) with a continuous duty voltage and current rating sufficient to assure adequate power for the application of tungsten carbide coatings of the thickness specified herein. The output shall have the voltage regulation required to prevent damage to the surface being coated. The output power shall be continuously variable, or adjustable throughout the output range of the power supply in order to increase or decrease the coating thickness. Connectors shall be provided for the cable connecting the power supply to the gun. The machine shall be completely wired through the controls and ready for connection to the power source. The power supply and control console shall be portable and the weight shall not exceed 50 pounds.

3.4.2 Controls and indicators. The machine shall be equipped with the following controls: An alternating current (AC) input power "on-off" switch and indicator lamps or an AC voltmeter to indicate when power is applied to the unit and output power controls in the form of switches or rheostats to control the output power. DC voltmeters and ammeters or calibrated controls shall be used to determine the power output.

3.4.3 Gun. The gun shall be designed and balanced for ease of handling during operation. A holder on the gun shall be provided for mounting the electrode, and shall accommodate electrode diameters within a range of at least 1/16 through 3/16 inch. The holder shall provide a firm electrical and mechanical connection between the holder and electrode. A vibrator shall be provided to vibrate the electrode axially at a frequency of at least 60 Hz. A switch shall be mounted at a convenient location on the gun to control power to the electrode and vibrator. The gun weight shall not exceed 2.5 pounds.

3.4.4 Electrical system. Unless otherwise specified (see 6.2.1), electrical input power shall be 115 volts single phase, 60 Hz., AC. The power cable and plug for connecting to the primary power source shall conform to NFPA 70.

3.4.5 Solid state components. Solid state design shall be used throughout for electronic components.

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3.5 Performance. The machine shall be capable of depositing tungsten carbide coatings at a rate of coverage of not less than two inches per minute over a surface 0.125 inches wide with the thickness requirements specified in table I. The coating shall not chip or flake off of the surface to which it is applied.

TABLE I. Requirements.

Size	Deposit thickness	
	minimum (in.) 1/	maximum (in.) 2/
2	0.0001	0.002
3	0.0001	0.004

1/ Thickness shall be not greater than value stated.
2/ Thickness shall be not less than value stated.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully decribed (see 6.2.1).

3.6 Standard equipment. Unless otherwise specified (see 6.2.1), standard equipment normally furnished by the manufacture shall be furnished.

3.7 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.9 Marking of instruction plates, indicating plates, and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.9.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)

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Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.11 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall

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consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be connected to an appropriate electrical power source and tested under actual operating conditions. The test shall consist of applying a tungsten carbide coating to an appropriate steel specimen. Any evidence of malfunction of the machine observed during application of the coating or by examination of the coating following application shall be cause for rejection.

4.7.2 Performance test. The machine shall be tested to determine conformance with 3.5 by coating a test speciman. Preparation of the speciman and the coating technique shall be the responsibility of the manufacturer and as described in their operating instruction manual. The speciman shall be coated to a predetermined thickness. The time required to apply the coating shall not exceed one minute. The coating shall be sufficiently dense to thoroughly cover the speciman, and the thickness shall be within plus or minus 0.0001 inch of the predetermined value. The coating shall be tested to assure that it is bonded to the surface and will not chip or flake off.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The machine described in this specification will be used primarily for coating the cutting surfaces of metal cutting tools. This coating is intended to reduce wear and increase tool life.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific measurement system (US or SI), state required system (3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Electrical system and input power, if different (see 3.4.4).

i. Additional performance features, if required, (see 3.5.1).

j. Standard equipment, if different (see 3.6).

k. Specify and fully describe optional equipment, if required (see 3.7).

l. Specify applicable repair parts, if required (see 3.8).

m. Nameplate, if different (see 3.9.1).

n. First article inspection, if different (see 4.4).

o. Quality conformance inspection, if different (see 4.5).

p. If preservation, packing, and marking in accordance with is MIL-W-45562 required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the con-

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tract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Coatings
Deposits
Electrodeposition
Electroplating

6.5 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

MIL-D-80229	MIL-D-80229A
Size 1	Cancelled
Size 2	Size 2
	Size 3 (New size)

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3433-0100)

Review activity:
Air Force - 84

User activities:
Army - AV
Navy - MC

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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